

2	Two spheres A and B approach each other as illustrated in Fig. 2.1. Before collision A: 0.500 kg, 7.40 m s⁻¹ (right) B: 0.350 kg, 9.60 m s⁻¹ (left) After collision A: 0.500 kg, 0 m s⁻¹ (at rest) B: 0.350 kg, 10.4 m s⁻¹ (right) Fig. 2.1 Sphere A has a mass of 0.500 kg and moves to the right with a speed of 7.40 m s⁻¹. Sphere B has a mass of 0.350 kg and moves to the left with a speed of 9.60 m s⁻¹. The spheres collide and are in contact for a time of 0.400 s. Sphere B reverses its direction of motion and moves off with a speed of 10.4 m s⁻¹.
	(a) Using momentum consideration, explain quantitatively why spheres A and B cannot be at rest at the same instant.
	
	

			[2]
L2	Take right as positive direction. Total initial momentum of A and B $= (0.500)(+7.40) + (0.350)(-9.60)$ $= +0.34 \text{ kg m s}^{-1}$ <u>No net external force acts on system of A and B, therefore total momentum is conserved at all times and always equal to the total initial momentum.</u> <u>Total initial momentum equals 0.34 kg m s^{-1} to the right, which is non-zero.</u> Therefore both A and B cannot be at rest at the same time.	M1 A0	
(b)	For the time during the collision, calculate the average force between the spheres.		
	average force = N		[2]
L2	Take right as positive direction. Considering sphere B's rate of change in momentum, $\text{Average force} = \frac{\Delta p}{\Delta t} = \frac{0.350(10.4 - (-9.60))}{0.400}$ $= 17.5 \text{ N}$ (i.e. to the right) <i>*Mark for magnitude only.</i>	M1 A1	
(c)	Use your answer in (b) to determine the magnitude of the velocity of sphere A after the collision. Explain your working.		
	magnitude of velocity = m s^{-1}		[3]
L2	Take right as positive direction. By Newton's 3rd Law, the force experienced by A is equal in magnitude and opposite in direction to the force experienced by B: Average force on A = -17.5 N $\text{Average force} = \frac{\Delta p}{\Delta t}$ (Average force on A) $\times \Delta t = \Delta p \text{ of A}$ $(-17.5) \times 0.400 = 0.500(V_A - 7.40)$ $V_A = -6.60 \text{ m s}^{-1}$ (i.e. to the left) B1 mark: Explanation of working (sign convention is clearly defined; application of Newton's 3rd law) M1 mark: Calculation (equation used and substitution of values are clear and correct) A1 mark: Correct answer for magnitude of velocity of A.	B1 M1 A1	

	(d)	By considering quantitatively the relative speeds of approach and of separation of the two spheres, deduce whether the collision is elastic or inelastic.	
			
			
			[2]
	L2	Take right as positive direction. Relative speed of approach = $U_A - U_B = 7.40 - (-9.60) = 17.0 \text{ m s}^{-1}$ Relative speed of separation = $V_B - V_A = 10.4 - (-6.60) = 17.0 \text{ m s}^{-1}$ Since the relative speed of approach is equal to the relative speed of separation, it implies that total kinetic energy before and after the collision is unchanged, hence it is an elastic collision. <i>Allow ECF from previous part.</i> <i>Calculation of relative speeds considered positive & negative directions of velocities.</i> <i>Conclusion based on comparison of calculated relative speeds.</i>	M1 A1

[Total: 9]

2	(c)	Copper has one conduction electron per atom. The density of copper is 8960 kg m^{-3} . The
---	-----	--